



BACHELOR OF COMPUTER SCIENCE · FINAL YEAR PROJECT

ClarityWorks

A Full-Stack Text Readability Analysis Platform using Machine Learning,
NLP, and Retrieval-Augmented Generation

Machine Learning

NLP

RAG

Full-Stack

React + Flask + Express

Junaid Malik · Wahaj Naveed · 2026

MOTIVATION

The Problem

Readability Matters

36% of adults read at or below a basic literacy level. Mismatched reading material leads to poor comprehension, disengagement, and inequitable access to information.

Existing Tools Fall Short

Traditional formulas like Flesch-Kincaid use surface metrics only. They cannot detect semantic complexity, passive voice density, or academic vocabulary.

No End-to-End Solution

Current tools either analyse *or* simplify — never both. Teachers, editors, and researchers need a single platform for analysis, rewriting, and reference material search.

OUR GOAL

Build a unified platform that **analyses**, **rewrites**, and **augments** text readability using ML, NLP, and Generative AI.

5K+

CLEAR CORPUS PASSAGES

16

ML FEATURES

3

ENSEMBLE MODELS

Three-Tier Microservice Design

Frontend

React 18 · TypeScript · Vite
Port 5173

- 15+ pages / 30+ components
- Recharts visualisations
- ReactFlow concept graphs
- TailwindCSS + custom design system
- PDF/DOCX export pipeline

REST API



Backend

Node.js · Express · TypeScript
Port 5000

- JWT authentication (bcrypt)
- PostgreSQL (5 tables)
- Role-based access control
- File upload (Multer)
- Proxy to ML microservice

HTTP



ML Service

Python · Flask · scikit-learn
Port 5001

- 3-model ML ensemble
- spaCy NLP pipeline
- Bidirectional rewrite engine
- ChromaDB + RAG
- LLM integration (Llama 3.3 70B)

PostgreSQL 14

ChromaDB Vector Store

E5-small-v2 Embeddings

FlashRank Re-ranker

ML-Based Grade Level Prediction

CLEAR Corpus Dataset

~4,724 professionally graded reading passages (Grades 3–12) from CommonLit & Georgia State University. Target variable: Flesch-Kincaid Grade Level.

16 Engineered Features

- Word Count
- Sentence Count
- Avg Sentence Length
- Avg Syllables/Word
- Difficult Words %
- Flesch Reading Ease
- FK Grade
- ARI
- SMOG
- Type-Token Ratio
- Avg Word Length
- Passive Voice %
- Subordinate Clause Density
- POS Diversity
- Lexical Diversity
- Sentence Variance

Blue = original features · Purple = spaCy NLP features (added for deeper linguistic analysis)

3-Model Ensemble

Random Forest

300 trees

Gradient Boosting

100 estimators

XGBoost

300 estimators

Prediction = average of all 3 models · Confidence from inter-model agreement

0.712

MEAN ABSOLUTE ERROR
(grade levels)

0.926

R² SCORE
(variance explained)

80.2%

WITHIN ±1 GRADE

Comprehensive Readability Analysis

5 Input Methods

- Direct text paste / type
- PDF upload with extraction
- DOC/DOCX file upload
- Image OCR (Tesseract)
- Voice transcription (Web Speech API)

Analysis Output

- ML grade prediction (Grade 3 – College)
- 8 readability formula scores
- Complexity Score (0–100 composite)
- Difficulty-adjusted reading time
- Difficult word & sentence detection
- Vocabulary level analysis
- Improvement suggestions with grade impact
- Concept prerequisite graph
- 6+ interactive chart visualisations

Analysis 10/05/2026

10/05/2026, 22:16:14

GRADE LEVEL

Grade 12

Advanced

FLESCH EASE

64.9

Fairly easy

WORD COUNT

347

11 sentences

DIFFICULT WORDS

7

2.0% of total

READING TIME

1 min 48 sec

Moderate reading pace - 193 WPM

Why Grade 12?

Technical

This text is very advanced, suitable for high school seniors (17-18 years old) preparing for college. It features college-level vocabulary and argumentation requiring critical analysis.

Vocabulary Level

College-preparatory, specialized

Sentence Length

26-30+ words per sentence

Yours: 31.6 words

Structure

College-level complexity and abstraction

Target Audience

High school seniors (ages 17-18)

Your Text Analysis:

Average 31.6 words per sentence

Average 1.34 syllables per word

2.0% difficult words

Flesch Reading Ease: 64.9 / 100

Text Complexity Score

Show Breakdown

57 /100

Data-Driven Insights

Readability Scores



5-axis radar showing Flesch, FK Grade, ARI, SMOG, and Coleman-Liau normalized to 0–100.

Vocabulary Level Distribution



Words categorised into Simple / Medium / Advanced / Expert with colour-coded distribution.

Text Complexity Score



Weighted composite 0–100: grade level (40%), Flesch (30%), difficult words (20%), sentence length (10%).



Detailed PDF Report

Multi-page jsPDF with cover, scores, suggestions, vocabulary



Heatmap Highlighting

Difficult words & sentences colour-coded with hover tooltips



Dashboard Trends

Grade & Flesch score trend line over last 20 analyses

CORE FEATURE

Bidirectional Text Rewriting Engine

Rewrite in Any Direction

Simplify text *down* to Grade 3 or upgrade *up* to College level. The engine auto-detects direction based on current vs. target grade.

Multi-Round Beam Search

- Beam width of 3 — tracks top 3 candidates
- Rounds: lexical → split/combine → discourse
- Grade-specific policy buckets control intensity
- Scoring: syllables/word + words/sentence distance

Synonym Cascade (5 Sources)

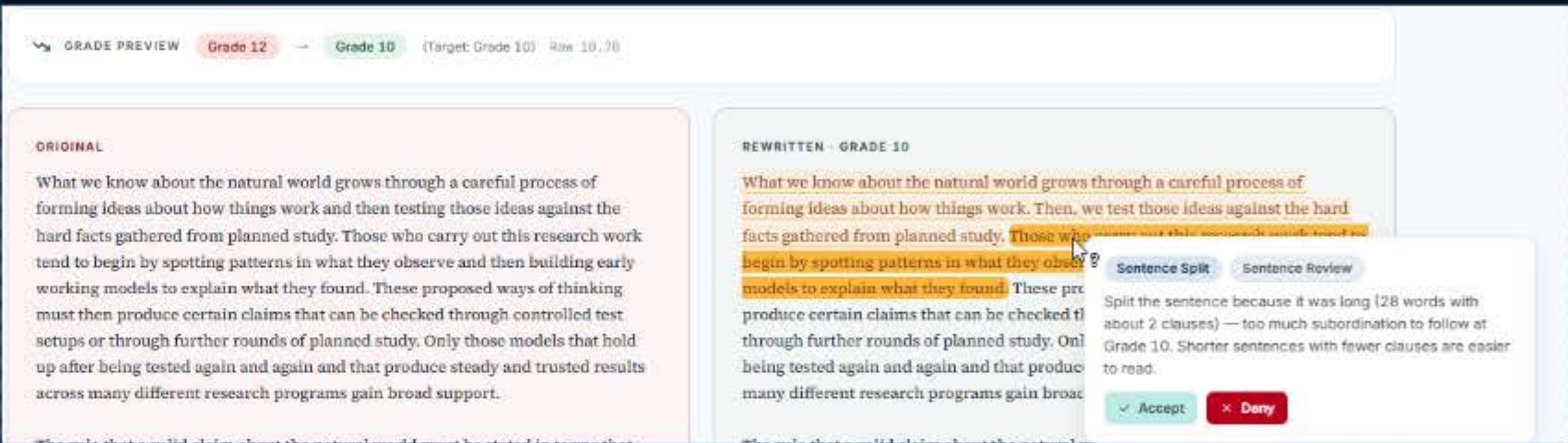
- 1 Curated simplification map (90+ pairs)
- 2 WordNet + Lesk WSD (context-aware)
- 3 Datamuse API (frequency-ranked)
- 4 LLM rescue (Llama 3.3 70B)
- 5 Complexification map (for upgrades)

LLM Validation & Repair (3-Pass)

Pass 1: Validate

Pass 2: Critic Review

Pass 3: Polish & Fix



Auto Mode

Best candidate applied automatically with full change log

Interactive Mode

Accept or deny each change individually with inline highlights

1

Retrieves top-20 candidates per collection from ChromaDB

2

FlashRank re-ranks to precise top-5 results

3

Llama 3.3 70B generates multi-paragraph answers with citations

- PDF (pdfplumber) & DOCX extraction
- Automatic embedding model migration
- Textbook-specific text cleaning (headers, footers, TOC removal)

SELECT SOURCES

☒ Maleeha.pdf

☒ Halima.pdf

☐ Al Sweigart - Automate the Boring Stuff with Python, 3rd Edition (Early Access) - No Starch Press (2024).pdf

Clear selection (search all)

Searching 2 document(s)

SEARCH QUERY

Tell me how we can use Microsoft Presidio for LLM prompt safety

Press Enter to search

Search

Answer

5 sources

Microsoft Presidio can be used for LLM prompt safety by detecting and redacting Personally Identifiable Information (PII) from text, thereby preventing potential leaks and attacks [Source 1]. This is achieved through a spaCy NLP pipeline against a registry of entity recognizers, with three custom recognizers added for specific types not in the default set.

The Presidio stage is specifically designed to catch PII and credential leakage, including API keys, identity numbers, and card details [Source 1]. It runs input through a pipeline and discards detections below a certain confidence threshold before results reach the policy engine. This process helps to filter out harmful or sensitive information from LLM prompts, reducing the risk of exposure under frameworks like GDPR [Source 1].

In the context of SecureLayer, a modular LLM security gateway, Microsoft Presidio is used as a PII scanner with three custom recognizers [Source 3]. This scanner is part of a three-stage process that includes a rule-based injection scoring engine and a policy engine, which assigns Allow, Mask, or Block to each request based on the presence of PII or other threats. The system has been tested across fifteen scenarios and achieved an accuracy of 93.3% at an average latency of 67 milliseconds [Source 3].

Furthermore, PromptGuard, another system that utilizes Microsoft Presidio, runs three stages in sequence, including a Threat Scorer, a PII Engine using Presidio, and an Action Engine [Source 4]. The PII Engine outputs a PIIReport, which is combined with the Threat Scorer output to issue a final verdict of Allow, Mask, or Block. This process helps to detect and prevent potential threats, including structural embedding attacks and PII leaks [Source 4]. The provided sources do not state the full range of customization options available for Microsoft Presidio, but they do highlight its importance in detecting and redacting PII from LLM prompts [Source 1, Source 3, Source 4].

Based on your uploaded textbooks. Expand source documents below to verify.

Choose PDF or DOCX File

Supports large textbooks (hundreds of pages). Processing may take a few minutes.

Uploaded Textbooks 3

CORE FEATURE

Concept Prerequisite Graphs

What knowledge does a reader need?

Concept graphs map the prerequisite knowledge structure of any text. They show what a reader must already understand before the main ideas will make sense.

Extraction Pipeline

- 1 **spaCy noun phrase extraction** — ranks top-30 key phrases by frequency
- 2 **LLM concept mapping** — Llama 3.3 70B identifies 5–15 concepts and prerequisite edges
- 3 **Validation & cleanup** — removes unconnected nodes, validates edge references

Three Concept Tiers

Prerequisite

Foundational knowledge

Intermediate

Bridging concepts

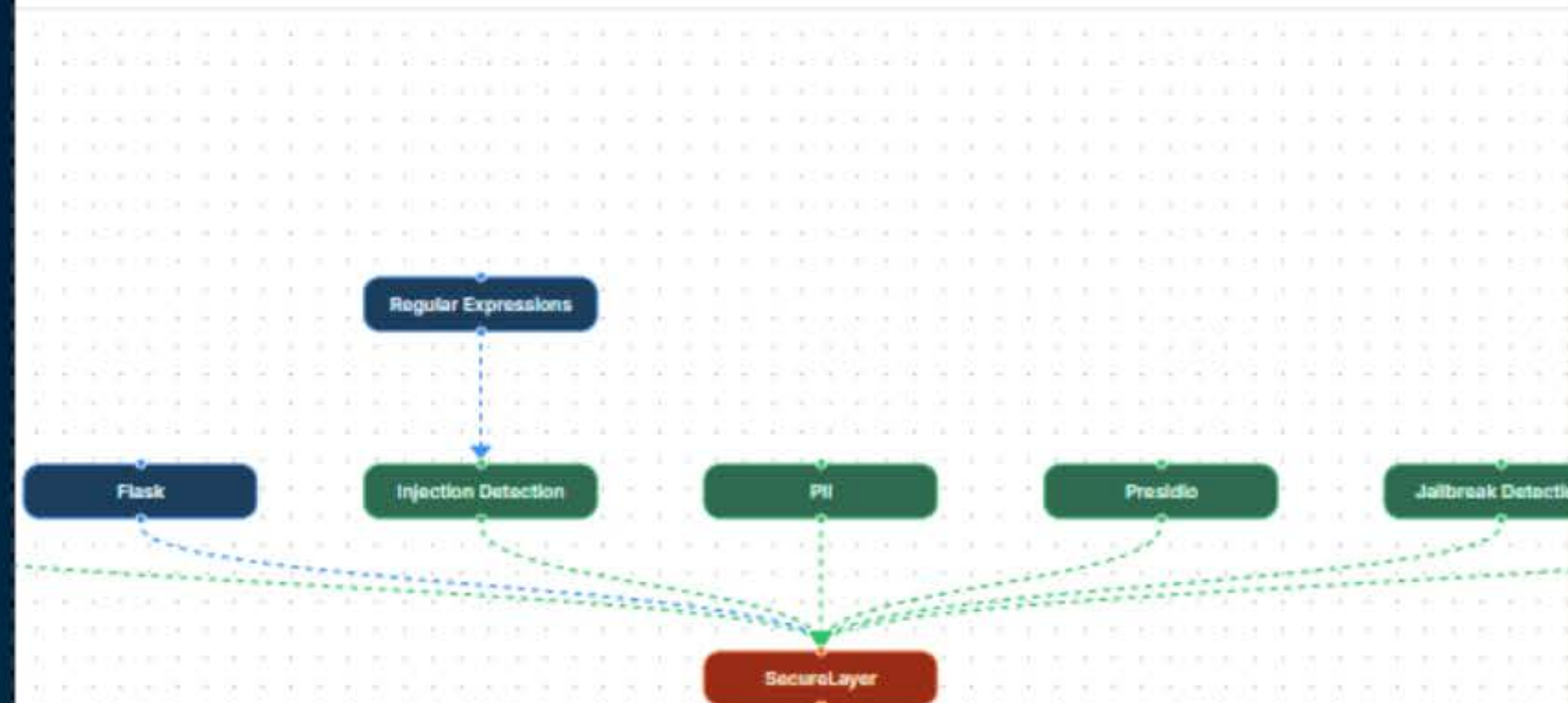
Target

Main ideas being taught

Available for both individual analyses and uploaded RAG documents.
Interactive drag-and-drop canvas with zoom controls.

Graph 10 concepts, 9 dependencies

Intermediate concept ■ target concept



Arrows mean "is required before". The graph shows what a reader must already know.

ADDITIONAL FEATURES

Compare, Batch, History & More

Compare

TEXT A

Grade 4

Beginner

View Full Analysis

TEXT B

College

Expert

View Full Analysis

Detailed Comparison

Metric	Text A	Text B	Diff
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Comparative Analysis

Analyse two texts side by side. Metrics comparison table with 11 dimensions, colour-coded deltas, and links to full individual analyses.

Results (9/9 successful)

Export CSV

#	Title	Grade	Flesch	Words	Sentences	Diff. Words %
1	Text 1	Grade 3	188.8	27	6	8.8%
2	Text 2	Grade 3	94.9	31	4	0.0%
3	Text 3	Grade 9	68.3	33	4	6.1%
4	The stock market experienced significant volatility during the thi...	College	61.8	19	2	52.6%
5	Scientists have long debated the epistemological implications of...	College	8.9	21	1	19.1%
6	Text 6	Grade 3	188.8	26	7	0.8%

Batch Processing

Paste multiple texts (separated by ---) or upload a CSV. Progress bar, summary statistics table, CSV export, clickable rows linking to individual analyses.

History

Paginated, searchable, filtered by grade

12 Sample Texts

Grade 3 to College, calibrated

Dark Mode

Full theme toggle, persisted

Admin Panel

User management, stats, RBAC

Technology Stack



Frontend

React 18	UI framework
TypeScript	Type safety
Vite 5	Build tool
TailwindCSS	Utility-first CSS
Recharts	Data visualisation
ReactFlow	Concept graphs
jsPDF	PDF generation
Axios	HTTP client



Backend

Node.js	Runtime
Express.js	REST framework
PostgreSQL	Relational DB (5 tables)
JWT + bcrypt	Auth & security
Multer	File uploads
TypeScript	Type safety



ML Service

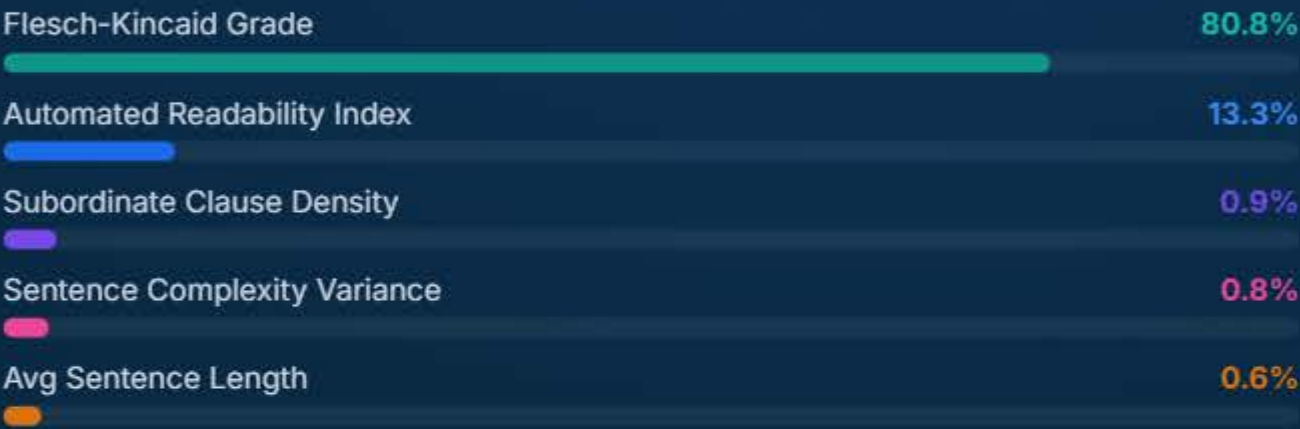
Flask	API framework
scikit-learn	RF, GB models
XGBoost	Ensemble model
spaCy	NLP pipeline
ChromaDB	Vector DB (RAG)
E5-small-v2	Embeddings
FlashRank	Cross-encoder reranker
Llama 3.3 70B	LLM (via Fireworks AI)

Testing & Results

ML Model Performance

METRIC	VALUE
Mean Absolute Error	0.712 grades
R ² Score	0.926
Within ±1 Grade	80.2%
Training Samples	~4,724

Feature Importance (Top 5)



Rewrite Engine Validation

TEST CASE	RESULT
Grade 3 → Grade 6	Predicts Grade 5 ✓
Grade 3 → Grade 8	Predicts Grade 9 ✓
Grade 3 → Grade 10	Predicts Grade 10 ✓

Bidirectional rewriting tested across downgrade and upgrade paths with semantic similarity > 0.82 threshold.

SUMMARY

ClarityWorks

A comprehensive platform combining traditional readability formulas, a 3-model ML ensemble, NLP-powered text rewriting, and RAG-based textbook search — all within a polished, modern web interface.

30+

REACT COMPONENTS

25+

API ENDPOINTS

7,500+

LINES (SIMPLIFIER)

0.926

R² ACCURACY

Thank you · [Questions?](#)

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